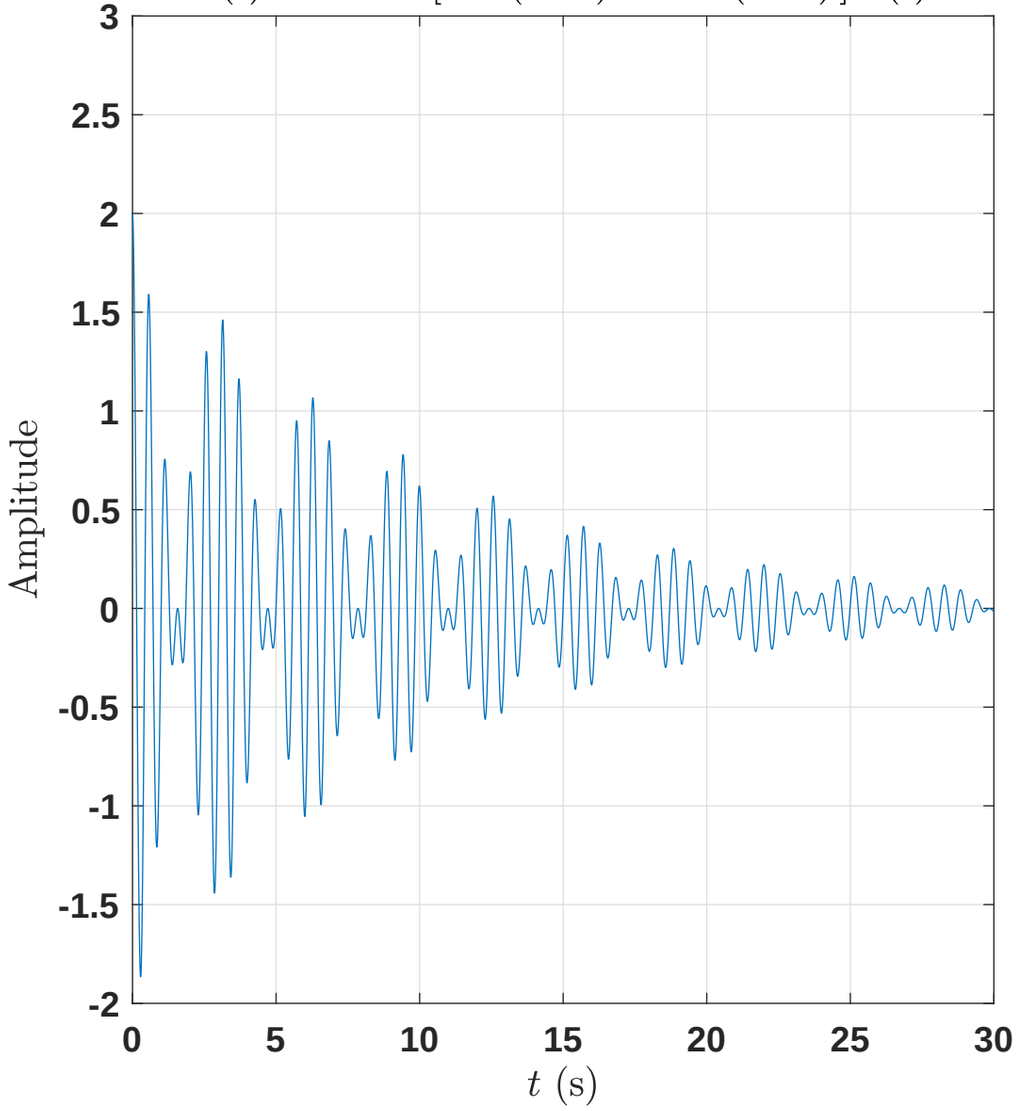
$$x(t) = e^{-0.1\,t}\,[\,cos\,(\,10\,t\,)\,+\,cos\,(\,12\,t\,)\,]\,u(t)$$



$$X(\omega_c) = \frac{0.1 + j\,\omega_c}{(\,0.1 + j\,\omega_c\,)^2 + 10^2} + \frac{0.1 + j\,\omega_c}{(\,0.1 + j\,\omega_c\,)^2 + 12^2}$$

